Rushil Srirambhatla

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Education

- **Johns Hopkins University** — Baltimore, MD
  B.S. Applied Mathematics & Statistics, Cognitive Science; GPA: 3.83/4.00 — *Aug 2023 – Dec 2026*
  - **Relevant Coursework**: Deep Learning, Artificial Intelligence, Linear Algebra, Probability, Mathematical Statistics, Mathematical Foundations of Data Science, Computer Vision, Optimization, Data Structures & Algorithms

Research Experience

- **Stanford University – AIDE Lab** — Stanford, CA
  Machine Learning Research Intern — *Jun 2026 – Present*
  - Built a scalable **3D CT preprocessing pipeline** for **25,489 Merlin studies**, generating **23,935 VAE embeddings** and segmentations using **TotalSegmentator**; implemented **SLURM** workflows for distributed GPU processing.
  - Fine-tuned three **text-conditioned 3D diffusion U-Nets** using **MAISI**, frozen **ClinicalT5** embeddings, and cross-attention; compared impression-only, full-report, and **MedGemma-27B**-structured report conditioning on matched training and validation cohorts.
  - Improved medical text-image alignment by **42%** (BiomedCLIP: **14.95 to 21.19**) using structured reports; identified systematic **512-token truncation** and redesigned prompts to preserve clinical impressions in **99.8%** of training cases.
  - Developed a multidimensional evaluation framework combining **FID, BiomedCLIP, Merlin-based clinical assessment, and 3D anatomical measurements**; conducted **74-patient fixed-noise counterfactual experiments**, identifying limitations in disease-specific anatomical control.

- **BIOS Clinical Trials Coordinating Center** — Remote
  Machine Learning Researcher — *May 2024 – Present*
  - Led a large-scale benchmark of **seven CT brain-extraction algorithms** across **5,062 scans** from **499 patients at 78 sites**, developing task-specific evaluation criteria for registration, volumetric analysis, and deep-learning workflows.
  - Developed and validated **Robust-CTBET**, integrating CT intensity windowing, template-guided neck removal, and spatial registration; achieved a **0.30% scan-level failure rate** and **Dice ≈ 0.98**, outperforming six competing methods.
  - Built reproducible **Python/R pipelines** for large-scale 3D CT preprocessing and quality control; evaluated methods using **Dice, HD95, MSD, ICV error, and rim-HU contamination**, with **bootstrap confidence intervals** and scanner-stratified statistical analyses.

- **University of Pittsburgh Medical Center** — Pittsburgh, PA
  Deep Learning Researcher — *May 2021 – Present*
  - Developed scalable **3D MRI preprocessing and brain-age prediction pipelines** across **40,841 scans from 46 datasets**, using **ANTs** registration, intensity normalization, and **3D CNNs** in **PyTorch Lightning**; achieved **3.14-year test MAE** and investigated disease-specific brain aging.
  - Developed a **skip-connection-free 3D autoencoder** with a **512 dimensional latent bottleneck** and **3D pixel-shuffle decoder** for whole-brain MRI reconstruction; reduced **GPU memory by 19.6%**, runtime by **10.4%**, and parameters by **13.4%**, evaluating fidelity using **MAE and SSIM**.
  - Developed a **WMH microstructure analysis pipeline** for **576 participants**, integrating **3D U-Net (LST-AI)**, **ANTs**, and T1w/T2w-FLAIR imaging with **multiple imputation and mediation analysis**.
  - Developed an explainable **CNN for COVID-19 CT classification** using **MATLAB**, HU-windowed lung isolation, and **class activation mapping (CAM)**, achieving a **12.34% improvement** over reference models.

- **Massachusetts General Hospital / Harvard Medical School** — Boston, MA
  Neuroimaging Researcher — *Mar 2025 – Present*
  - Developed a reproducible **Python/R neuroimaging analysis pipeline** integrating cognitive assessments and **FreeSurfer-derived cortical thickness** across **68 brain regions** in the ADNI cohort.
  - Applied **age-stratified multivariable regression, ANCOVA, and FDR correction** to examine cortical structure and cognitive performance in **cognitively healthy older adults**.
  - Identified stronger associations between **executive function and cortical thickness** in adults aged **80+**, with widespread **FDR-significant regional associations** across temporal, parietal, frontal, and occipital cortices.

- **Washington University School of Medicine** — St. Louis, MO
  Computational Cancer Genomics Researcher — *May 2025 – Dec 2025*
  - Developed **machine-learning screening models for IDH1/2-mutant clonal hematopoiesis** using **457,698 UK Biobank participants**, combining FDR-controlled feature selection with elastic net, random forest, and XGBoost classifiers.
  - Achieved **56.9-fold enrichment** in mutation-carrier prevalence among the top **0.1% of predicted-risk participants** on held-out data, supporting targeted genetic screening.

Publications

- **Srirambhatla, R.**; Walborn, N.; Hanley, D.; Muschelli, J. *Benchmarking Automated CT Brain Extraction Methods Across 5,000 Scans.* BMC Medical Imaging, pending acceptance.

- **Srirambhatla, R.**; Karim, H. T.; Campion, J.-Y. *Memory-Efficient Skip-Connection-Free 3D Autoencoder for Whole-Brain MRI Reconstruction.* ECCV 2026 AI4M3D Workshop, accepted.

- Jain, R.; **Srirambhatla, R.**; Kessler, J.; Goel, R. *The Black Box Dilemma: Challenges in Human-AI Collaboration in ML-CDSS. The American Journal of Bioethics*, 24(9), 108–110, 2024. Published commentary. doi:10.1080/15265161.2024.2377105.

- **Srirambhatla, R.**; Campion, J.-Y.; Desmidt, T.; Pan, Y.; Andreescu, C.; *et al. Altered T1w/T2w-FLAIR Ratio in White Matter Hyperintensities as an Indicator of Structural Integrity Loss: Association with Alzheimer's Disease and Vascular Dementia.* medRxiv preprint, 2026; under revision. doi:10.64898/2026.07.28.26359139.

- Campion, J.-Y.*; **Srirambhatla, R.***; Andreescu, C.; *et al. Deep Learning-Derived Brain Age from 40,000+ MRI Scans Reveals Disease-Specific Aging Trajectories.* Submitted. *Equal contribution (co-first authors).

- **Srirambhatla, R.**; Hong, Y.; Zhang, Y.; Salat, D. H. *Cortical Thickness Signatures of High vs. Low Cognitive Performers in the ADNI Dataset.* medRxiv preprint, 2026.

- **Srirambhatla, R.**; Karim, H. T. *Prediction of COVID-19 Diagnosis from Healthy and Pneumonia CT Scans Using Convolutional Neural Networks.* medRxiv preprint, 2022. doi:10.1101/2022.10.20.22281334.

Conference Presentations & Abstracts

- **Srirambhatla, R.**, Walborn, N., Hanley, D., & Muschelli, J. *Comparison of Automated Brain Extraction Methods in Over 5,000 Computed Tomography Scans.* International Stroke Conference, 2026. *Stroke*, 57(Suppl. 1), WP305. doi:10.1161/str.57.suppl_1.WP305.

- **Srirambhatla, R.**, *et al. Altered T1w/T2w Ratio in White Matter Hyperintensities as an Indicator of Structural Integrity Loss.* Alzheimer's Association International Conference (AAIC) and Society of Biological Psychiatry (SOBP), 2026.

- Campion, J.-Y., **Srirambhatla, R.**, Andreescu, C., Desmidt, T., & Karim, H. T. *Deep Learning-Derived Brain Age from 40,000+ MRI Reveals Disease-Specific Aging Trajectories.* Alzheimer's Association International Conference (AAIC) and Society of Biological Psychiatry (SOBP), 2026.

- **Srirambhatla, R.**, Mustion, G., Tran, D., Chatterjee, N., & Bolton, K. *Machine Learning–Driven Screening for Individuals at High Risk of IDH1/2-Mutant Clonal Hematopoiesis.* 15th International Workshop on Molecular Aspects of Myeloid Stem Cell Development & Leukemia, 2026.

- **Srirambhatla, R.**, Hong, Y., Zhang, Y., & Salat, D. H. *Cortical Thickness Signatures of High vs. Low Cognitive Performers in the ADNI Dataset.* Poster presentation, Mass General Brigham Neuroscience Day, 2025.

- **Srirambhatla, R.** & Karim, H. T. *A Task-Specific Curriculum Learning Approach to Predict the Onset of Alzheimer's Disease.* Johns Hopkins DREAMS, 2024.

- Bagga, A.; Bharti, S.; Polamraju, A.; Shah, M.; Edupadanti, H.; **Srirambhatla, R.**; *et al. Persistent Insurance Coverage Gaps Among Working-Age United States Stroke Survivors: No Improvement From 1999 Through 2023.* International Stroke Conference, 2027. Oral presentation accepted.

Honors, Awards & Fellowships

- **Stanford Radiological Sciences Laboratory (RSL) REU** – $8,000 (2026)
- **WashU Computational Cancer Genomics Summer Research Program** – $7,000 (2025)
- **JHU Bloomberg Distinguished Professors Summer Program** – $6,000
- **Best Delegate** – UNC Model UN Conference (2024)
- **Dean's List** – 2024, 2025, 2026
- **Northwest Science Expo** – $9,000/year renewable OSU scholarship
- **Biomedical Hackathon** – 1st Place (Med-O-Vate)

Leadership, Community Service & Clinical Experience

- **Founder, Welfare Support Organization** – Donated 1,000+ lbs of clothing to 120+ homeless individuals (100+ hours).
- **Clinical Volunteer, MedStar Health** – Patient transport, vital signs, and clinical support (25+ hours).
- **Volunteer, Alzheimer's Buddies & Elder Care** – Dementia care and patient companionship (15+ hours).
- **Physician Shadowing** – Observed pediatric neurosurgeries with Dr. Cohen and Dr. Lafaro (20+ hours).

Technical Skills

- **Programming & Systems** – Python, R, MATLAB, Bash, Java, Swift; Git, Docker, Slurm/HPC, LaTeX.
- **Machine Learning & AI** – PyTorch, PyTorch Lightning, Keras, TensorFlow, OpenCV, scikit-learn, TensorBoard, Matplotlib.